

**UNIVERSITY OF LIMERICK
OLLSCOIL LUIMNIGH**

FACULTY OF SCIENCE & ENGINEERING

**DEPARTMENT OF ELECTRONIC & COMPUTER
ENGINEERING**

MODULE CODE: ET4132 / ED5081

MODULE TITLE: Introduction to Web and Database Technology

SEMESTER: Autumn 2023-24

LECTURER: Professor Hussain Mahdi

INSTRUCTIONS TO CANDIDATES:

- Answer **All THREE Questions**: **Q(A)** consists of 26 multiple-choice questions and is worth **26 Marks**, **Q(B)** is worth **34 Marks** and **Q(C)** is worth **40 Marks**.
- The following **Negative marking scheme** is applied to **Q(A)**: **Correct Answer = 1 Mark**, **Incorrect Answer = – 0.25 Mark**, and **No Answer = 0 Mark**.
- This Exam is worth **40%** of the Module's total assessment. The remaining **60%** comes from your **Coursework**.

Q (A)

(34 Marks)

This Question comprises 26 multiple-choice Sections. You are required to answer All Sections. Each question is worth 1 Mark. **Negative marking** is applied: **1 Mark for a Correct Answer, – 0.25 Mark for an Incorrect Answer, and 0 Mark for a No Answer.**

Write your answers using, for example, this style: **Q(A):** 1. (b), 2. (d), 3. (a),

1. What is the correct XHTML tag for inserting a horizontal line ? (a) <he /> (b) <hl /> (c) <hr /> (d) <ho/>	8. If you put a <title> tag in the head section of an HTML document, its content will appear in bold, large print at the beginning of your page? (a) True (b) False (c) It depends on the browser (d) Only if the CSS makes it display that way
2. The attribute-value pair colspan= "2" can be added to only what HTML tag? (a) <table> (b) <tr> (c) (d) <td>	9. Which of the following is the correct HTML tag/code to define a list that lists the items with 1, 2, 3...? (a) <ol type="numerals"> (b) <ul type="num"> (c) (d) <ul type="1">
3. Which of the following is the correct use of the HTML <title> tag? (a) <title="Web Designer Help"></title> (b) <title=Web Designer Help/> (c) <title="Web Designer Help"/title> (d) <title>Web Designer Help</title>	10. What is the correct XHTML tag for adding a pre-selected checkbox to a form? (a) <input type="checkbox" name="animal" value="Dog" selected /> (b) <input type="checkbox" name="animal" value="Dog" selected="selected" /> (c) <input type="checkbox" name="animal" value="Dog" checked="checked"> (d) <input type="checkbox" name="animal" value="Dog" checked="checked" />
4. What are the <div> tags used for in HTML: (a) To replace paragraphs, i.e. to replace <p> tags (b) To logically divide paragraphs (c) To logically divide an HTML document (d) To provide space between paragraphs	11. What is the correct XHTML code for adding a one-line text field to a form? (a) <input type="text" /> (b) <input type="text"> (c) <textarea size="1"> (d) <text type="textfield" />
5. What is the correct HTML tag for inserting an e-mail link? (a) (b) <a mail href="xxx@yyy"> (c) (d) <a mail>xxx@yyy</mail>	12. What is the correct XHTML code for a scrolling list which allows multiple options selection? (a) <select name="cars" multiple> (b) <list name="cars" multiple /> (c) <select name="cars" multiple="multiple"> (d) <select name="cars" multiple />
6. Which of these tags are all <table> tags? (a) <table><tr><tt> (b) <table><head><tfoot> (c) <thead><body><tr> (d) <table><tr><td>	13. What is the correct tag for adding a field to an HTML form that allows multi-line text insertion? (a) <input type="text" rows="4" /> (b) <textarea> (c) <input type="text" size="4"> (d) <select type="text" size="4">
7. The tag has two required attributes: (a) href and alt (b) src and align (c) src and alt (d) type and src	14. What is the correct XHTML for inserting an image? (a) <image src="image.gif" alt="MyImage" /> (b) (c) image.gif (d)

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15. Which of the following entity names do you insert to display the '>' character in your webpage: (a) &lt; (b) &lt; (c) &gt; (d) gt&	22. The hierarchical database model deal with M: M relationships by: (a) splitting them into two relationships (b) introducing duplicate data and redundancy (c) creating many related tables (d) introducing tables with many rows but few columns
16. The correct HTML code for adding a background image to a web page using an in-line style is: (a) <body background= "sky.gif"> (b) <body image="sky.gif"> (c) <body style= "background-image: url('sky.gif')"> (d) <body style= "background-image: sky.gif">	23. Which one of the following real-world situations cannot be easily implemented using the hierarchical model: (a) each member can book many classes and each class can be booked by many members (b) each customer is subject to many transactions (c) each account at a bank can be either debited or credited (d) each college can have many members of staff
17. Where in an HTML document is the correct place to refer/link to an external style sheet? (a) In the <body> section (b) At the end of the document (c) At the top of the document (d) In the <head> section	24. In Relational database terminology, a field or a set of fields in a table that matches the primary key of another table is known as: (a) Master Key (b) Index Key (c) Foreign Key (d) Candidate Key
18. Which of the followings style rules uses a correct CSS syntax? (a) h1 {color: red; font-size: 18 pt;} (b) h1 {color: red; font-size: 18pt;} (c) h1 (color= red; font-size= 18pt) (d) {h1: color= red; font-size= 18pt;}	25. With SQL, how can you return the number of records in the "Staff" table? (a) SELECT COLUMNS(*) FROM Staff; (b) SELECT COUNT() FROM Staff; (c) SELECT RECORDS(*) FROM Staff; (d) SELECT COUNT(*) FROM Staff;
19. What is the correct style rule to define the colour and centre-align the text of all paragraphs in an HTML document ? (a) p {color: blue; text-align: center;} (b) p {colour: blue; align: center;} (c) p.center {color: blue; text-align: center;} (d) p {color= blue; text-align= center;}	26. With SQL, how can you delete the records where the "FirstName" is "Peter" in the Customers table? (a) DELETE RECORDS FirstName='Peter' FROM Customers (b) DELETE ROWS FirstName='Peter' FROM Customers (c) DELETE FROM Customers WHERE FirstName = 'Peter' (d) DELETE FirstName='Peter' FROM Customers
20. Which of the following DBMS is most appropriate for multi-user applications: (a) Lotus Approach (b) Ingres (c) File Maker Pro (d) Microsoft Access	
21. Which normal form requires that a table or tables in a relational database should have no transitive dependencies? (a) 1st Normal Form (b) 2nd Normal Form (c) 3rd Normal Form (d) 1st and 2nd Normal Forms	

Q (B)

- a) State whether the following SQL query is correct or incorrect and why. **If incorrect, provide the correct SQL code:** (9 Marks)

```
SELECT city, COUNT(student_id)
FROM Students
WHERE COUNT(student_id) > 5
GROUP BY city;
```

- b) The following tables form a relational database for registration information for a small university:

Students	(<i>student_id</i> , student_name, address, city, gender)
Lecturers	(<i>lecturer_no</i> , lecturer_name, phone, salary)
Courses	(<i>course_no</i> , course_name, department)
Modules	(<i>module_code</i> , <i>course_no</i> , lecturer_no, number_of_students)
Grades	(<i>student_id</i> , <i>module_code</i> , <i>course_no</i> , grade)

where,

- **Students** contains information on all enrolled students and *student_id* is the primary key.
- **Lecturers** describes the lecturers currently at the university and *lecturer_no* is the primary key.
- **Courses** lists information on all courses offered and *course_no* is the primary key.
- **Modules** describes each module of each course, and has the composite primary key: *module_code*, *course_no*.
- **Grades** contains records for each student's grade for every module of every course, and the composite key: *student_id*, *module_code*, *course_no* is its primary key.

Write suitable SQL codes to produce the followings (state any assumptions you make):

- (i) Create the Lecturers table. (6 Marks)
- (ii) Give Adam Smith (lecturer) a 5% pay salary increase on his annual salary plus €1,000 housing allowance. (5 Marks)
- (iii) A list of average grades and number of students for each module that is taken by ten or more students. Order the list alphabetically by module-code. (7 Marks)
- (iv) Total number of modules taught by lecturer Adam Smith. (7 Marks)

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Q(C)

- a) Identify and correct the mistakes, including any missing elements, in the following HTML code, using recommended HTML5 best practices and coding conventions we have used in the this module. For each error, your answer should provide the line number of the erroneous code, a very short description of the error, and the correct HTML code for that line. (20 Marks)

Line Number

```

1      <!DOCTYPE>
2      <html>
3          <title>My Webpage</title>
4          <head>
5          </head>
6          <style>
7              body {background-color= "aadd88";}
8              hr   { height: 12;}
9          </style>
10
11      <body>
12          <H2>My Heading Goes here</H2>
13          <p> This page has a number of mistakes that you need to identify
14          and suggest fixes for. </br>The page should conform to XHTML
15          Strict conventions</p>
16          <comment> Here's a horizontal line </comment>
17          <hr style="height:80px"  />
18          <p> Below is an example of a table and a drop down list</p>
19
20          <table style= width:280px>
21              <tr>
22                  <td>Item 1, Row 1</td>
23                  <td>Item 2, Row 1</td>
24              </tr>
25          </table>
26
27          <form action="mail: me@dom.net"  method="get">
28              <select>
29                  <option />First Item
30                  <option />Second Item
31                  <option />Third item
32              </select>
33          </form>
34      </html>

```

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- b) **Table.1** lists, in a flat-file format, Suppliers-Parts data for a given company specialised in PC assembly. The database stores data about parts which the company needs to purchase and the suppliers who can supply them. For each part, the list shows Part Number, Description, Manufacturer Number, Manufacturer Name and Country of origin. For each supplier, Supplier Number, Name and Address are given. For each part, the list also shows the quantity that can be supplied by each supplier.

Show and describe how to fully normalise **Table.1** by applying the 1st, 2nd and/or 3rd Normal forms as required by the Normalization process. Your answer should show the structures of the resulting tables at each stage, with their primary keys and foreign keys marked clearly.

(20 Marks)

Table.1: Suppliers-Parts data

PartNo	Description	ManufNo	ManufName	Country	SuppNo	SuppName	SuppAdd	Quantity
P1	motherboard	M1	Toshiba	Japan	S1	Jones	Dublin	200
		M2	Dell	USA	S2	Smith	London	300
		M3	HP	Scotland	S4	Ryan	Limerick	150
P2	sound card	M3	HP	Scotland	S1	Jones	Dublin	200
		M4	Hitachi	Japan	S3	O'Brien	Cork	300
		M3	HP	Scotland	S4	Ryan	Limerick	200
P3	power supply	M3	HP	Scotland	S2	Smith	London	300
		M4	Hitachi	Japan	S3	O'Brien	Cork	350

PartNo: Part Number
 Description: Description of the Part
 ManufNo: ID Number of the Manufacturer
 ManufName: Name of the Manufacturer
 Country: Manufacturer's Country of Origin
 SuppNo: ID Number of Supplier
 SuppName: Name of Supplier
 SuppAdd: Address of Supplier
 Quantity: Quantity of the Part that can be supplied by a given Supplier

END OF PAPER